

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459661.8399 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1909 +/- 0.0081
Transit depth (Rp/Rs)^2	3.64 +/- 0.31 [%]
Orbital Inclination (inc)	89.73 +/- 0.85
Airmass coefficient 1 (a1)	1.237 +/- 0.019
Airmass coefficient 2 (a2)	-0.177 +/- 0.025
Scatter in the residuals of the lightcurve fit is	1.58 %
Best Comparison Star	None
Optimal Aperture	4.11
Optimal Annulus	11.79
Transit Duration (day)	0.1029 +/- 0.0023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1909 ± 0.0081 vs 0.1457 (deviation 5.58σ)
Duration fit vs published	0.1029 d vs 0.12 d (deviation 7.43σ)
Mid-time O–C	-4.6 min
Depth SNR	23.6
Residual scatter	1.58 %

Light curve

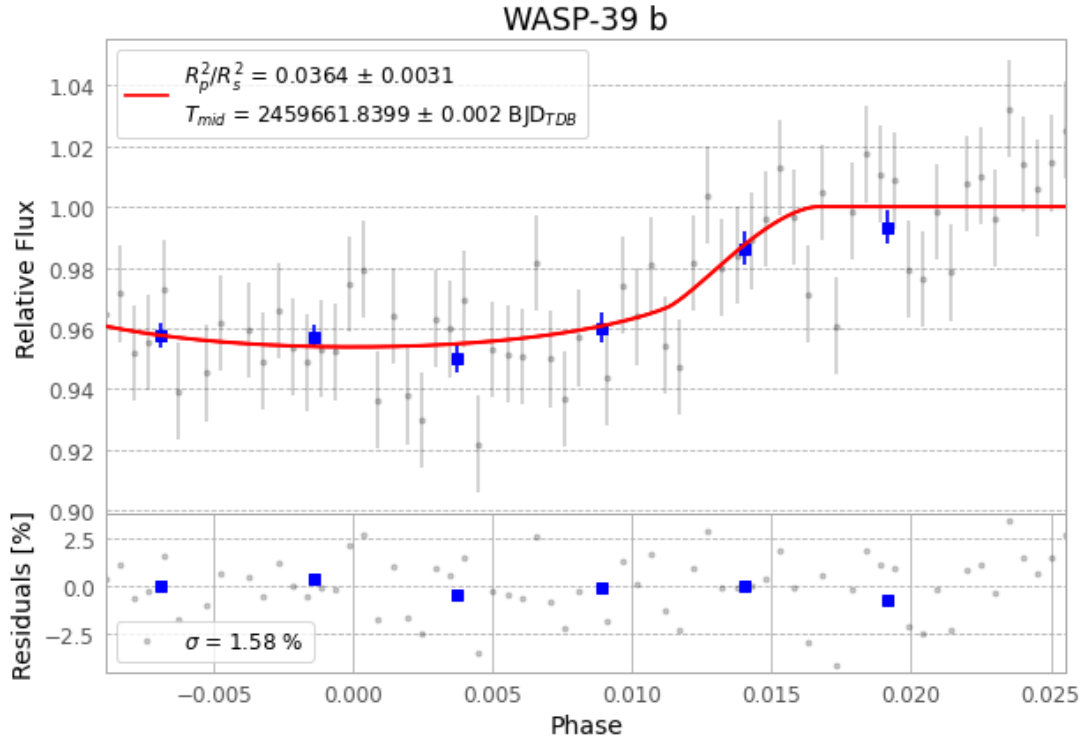


Figure 1 — FinalLightCurve_WASP-39 b_2022-03-23.png (SHA-256 2bee08e1dd84dcb9...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [198, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8c84442499a3f118...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), WASP-39220323070912.FITS → WASP-39220323103012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-220323.log (1ebe26372d30...), FinalLightCurve_WASP-39 b_2022-03-23.png (2bee08e1dd84...), FinalLightCurve_WASP-39 b_2022-03-23.csv (d40ba9827e34...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 17:04Z.